

Gourab Roy

Vision AI and Explainable AI Researcher | Erasmus Mundus IPCVAI

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Portfolio | GitHub | LinkedIn | Google Scholar

Profile

Erasmus Mundus IPCVAI master's student working on computer vision, model explainability, biomedical imaging, and safety-focused detection. Experienced in reproducible PyTorch experiments, SHAP faithfulness evaluation, YOLO inference, research writing, and student mentoring. Seeking a 5-6 month computer vision internship from February 2027, preferably in France and open to opportunities across Europe.

Education

Erasmus Mundus Joint Master's Degree - IPCVAI

Europe

Artificial Intelligence for Image Processing and Computer Vision

2025-2027, ongoing

- **University of Bordeaux, France - Semester 3, Sep 2026-Present:** Explainable AI for Images, AI and Sustainability, Advanced Computer Vision, Acquisition, Reconstruction and Medical Imaging, and TRDP 2.
- **Universidad Autonoma de Madrid, Spain - Semester 2, completed Jun 2026:** HPC for Deep Learning, Visual Signal Processing, Deep Learning for Computer Vision, Multi-view and Moving-camera Vision, and TRDP 1.
- **Pazmany Peter Catholic University, Hungary - Semester 1, completed Jan 2026:** Image Processing, Data Mining and ML, Deep Learning, Sensor Fusion, Biomedical Signal Processing, and Design Patterns.

Sylhet Engineering College

Sylhet, Bangladesh

B.Sc. in Computer Science and Engineering; CGPA: 3.56/4.00

2018-2023

- Thesis: *CerebralNet*, a deep learning workflow for detecting major abnormalities in brain CT and MRI using segmentation and classification.

Research and Teaching Experience

Sylhet Engineering College, Department of CSE

Sylhet, Bangladesh

Research Assistant

Nov 2023-Oct 2025

- Led and supported model design, evaluation, and paper preparation across medical AI, document AI, NLP, agricultural vision, and structural inspection, contributing to 12 peer-reviewed works.
- Mentored undergraduate students on final-year theses and research projects under Prof. Md. Abu Naser Mojumder.

Sylhet Engineering College, Department of CSE

Sylhet, Bangladesh

Adjunct Lecturer

Jan 2024-Aug 2025

- Taught Distributed Computing, Computer Architecture, Introduction to Data Science, Peripheral Interfacing, and Microprocessors and Microcontrollers.
- Designed assessments, held consultation sessions, and co-supervised final-year thesis projects.

Current Research

BERT vs CLIP with Whole-Word Partition SHAP

TRDP 2, Bordeaux, 2026

- Built a gated, resumable experiment comparing a fine-tuned BERT classifier with a frozen zero-shot CLIP text encoder on the same 500 balanced SST-2 sentences, producing 1,000 model-sentence explanation records.
- Evaluated classification, SHAP additivity, deletion faithfulness, lexical rationale proxies, bootstrap confidence intervals, and cross-model agreement. BERT achieved 0.936 macro-F1 versus 0.702 for CLIP; all 1,000 explanations passed additivity checks.
- Found limited explanation agreement between models (mean Spearman $\rho = 0.237$) and stronger SHAP-ranked deletion evidence than random deletion, especially for BERT.

Selected Projects

- SiteGuard AI - Real-Time PPE DetectionTeam software project, 2026
 - Trained and packaged a YOLO26n detector for person, helmet, vest, no-helmet, and no-vest classes using transfer learning; held-out test results: precision 0.906, recall 0.835, mAP50 0.878, and mAP50-95 0.592.
 - Developed image/video inference, worker-to-PPE association, structured JSON outputs, visualization, checksums, a model card, and 27 automated tests; documented domain-gap and temporal-smoothing limitations.
- Cardiac Autonomic Assessment via PPG-Derived HRVPPKE, 2025
 - Built a modular Python pipeline for 10 participants using Butterworth filtering, adaptive peak detection, FFT analysis, Friedman/Wilcoxon tests, and an individual responsiveness classifier.
- Calibrated Multi-View 3D ReconstructionUAM, 2026
 - Implemented camera calibration, SIFT/SURF/ORB matching, RANSAC verification, epipolar geometry, sparse reconstruction, and bundle adjustment in MATLAB; retained 672 tracks at 4.47 px reprojection error.

Selected Publications

- “CovidExpert: A Triplet Siamese Neural Network framework for the detection of COVID-19,” *Informatics in Medicine Unlocked*, Elsevier, 2023. DOI
- “Image-based classification of Bangladeshi rice varieties using deep convolutional neural networks,” *Agricultural Products Processing and Storage*, Springer Nature, 2026. DOI
- “PlantXNet: A CNN-Based Explainable AI Framework for Multi-Plant Leaf Disease Classification Using LIME,” *IEEE QPAIN*, 2026. DOI
- “MaizeXNet: A CNN-Based Explainable AI Framework for Maize Pest and Disease Classification,” *IEEE QPAIN*, 2026. DOI
- “PoultryXNet: An Explainable Deep Learning Framework for Poultry Disease Classification From Fecal Images,” *IEEE QPAIN*, 2026. DOI
- “Deep Learning-Based Detection of Aircraft Cracks and Dents Using YOLOv11 and YOLOv12 for Aviation Safety in Bangladesh,” *IEEE ICCCNT*, 2025.

Full publication list: [portfolio publications page](#) and [Google Scholar](#).

Technical Skills

Programming	Python, MATLAB, C, C++, Java, JavaScript
ML and Vision	PyTorch, TensorFlow, Scikit-learn, CNNs, Vision Transformers, YOLO, CLIP, BERT, transfer learning, segmentation, object detection, multi-view geometry
Explainability	Partition SHAP, attention rollout, deletion-based faithfulness, rationale proxies, additivity checks, model-agreement analysis
Scientific Stack	NumPy, Pandas, SciPy, Matplotlib, Jupyter, OpenCV, Hugging Face Transformers
Research Workflow	Git, pytest, LaTeX, model cards, experiment manifests, academic writing, peer review

Additional Information

Languages	Bangla (native), English (professional working proficiency)
Recognition	Reviewer for <i>Informatics in Medicine Unlocked</i> ; HackerRank gold badges in C++ and Python; 350+ programming problems solved
Availability	5-6 month computer vision internship from February 2027; France preferred, open to Europe